AI in Financial Risk Modelling

Can AI produce valid risk forecasts?

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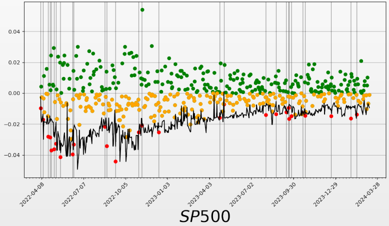
Where this course starts



—1% VaR threshold ● return up ● return down, above the threshold ● breach

- Pele et al. (2026) put a language model *inside* the risk number.
  - The history is written out as text, the model writes many possible tomorrows.
  - Value at Risk (VaR) is read off those draws.

The research question

The Thinker and the Sitting Woman

Hamangia culture, c. 5000 BC

- **Can AI produce a *valid* risk forecast?**
  - Getting an answer is easy: every model returns a number that looks like a VaR.
  - Basel counts breaches and capital follows the count (Basel Committee on Banking Supervision, 2019), so the answer has to hold up.

Valid at level α: conditional coverage

- $\mathbb{P}\Big(r_t < -\widehat{\mathrm{VaR}}_t(\alpha) \mid \mathcal{I}_{t-1}\Big) = \alpha$
- The rate is right, and the breaches do not cluster.

Four ways to produce tomorrow's 1% VaR

- **Classical benchmarks**, fitted on this asset alone.
  - Sort the window and read the quantile off it.
  - Or model the variance and impose a tail shape.
- A **time-series foundation model**, pretrained elsewhere.
  - It reads the price history and samples the next value.
- A **commercial language model**, handed the returns as text.
  - VaR comes out of prompted generation.
- An **open-weights language model** you run yourself.
  - The same prompt, but the weights are fixed and inspectable.

One question decides all four

- Can you tell, *from the output alone*, that the model answered the question you asked?

Outline

1. **Act I — What is a risk forecast?**
   - The functional, and the two factors it splits into.
   - Why 1% is not a smaller 5%.
2. **Act II — The classical answers.**
   - Historical simulation, GARCH, quantile regression.
   - What each assumes, and where it bites.
3. **Act III — Time-series foundation models.**
   - Zero-shot: pretrained elsewhere, used as it arrives.
4. **Act IV — Language models as risk forecasters.**
   - The prompt as a modelling decision.
   - What a valid-looking reply can still hide.
5. **Act V — Validation.**
   - Ranking, coverage and power.
   - Each one answers a different question.

By the end of the session, you should be able to. . .

1. Distinguish a **point forecast**, **marginal quantiles** and a **predictive distribution**.
   - Three different objects.
   - Only one of them is a threshold at *your* level.
2. Check that a model can produce the **functional you asked for**, before reading its numbers.
   - Ask it for several levels.
   - Then test whether the answers actually differ.
3. Separate **ranking**, **validity** and **power**.
   - Which forecast scores lower, and whether either one is valid.
   - Then whether the test could have told the difference.
4. Design an **as-of-valid** experiment with text and a pretrained model.
   - Every input dated before the forecast was made.
   - And a control that differs from it in one thing only.

Act I

What is a risk forecast?

A risk model targets a different object

- A **forecasting** model targets the centre of tomorrow's distribution; a **risk** model targets a quantile of it.

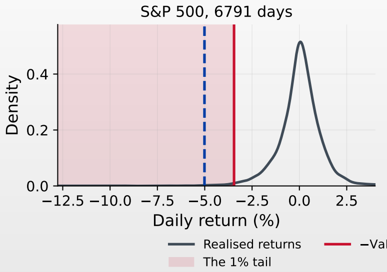
$$\hat{r}_t = \mathbb{E}[r_t \mid \mathcal{I}_{t-1}], \qquad \mathbb{P}(r_t < q_t(\alpha) \mid \mathcal{I}_{t-1}) = \alpha$$

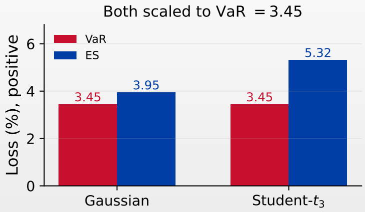
- With $L_t = -r_t$ the loss and $q_t(\alpha) < 0$ the return quantile, Value at Risk and Expected Shortfall (ES) are:

$$\mathrm{VaR}_t(\alpha) = -q_t(\alpha) > 0, \qquad \mathrm{ES}_t(\alpha) = \frac{1}{\alpha}\int_0^\alpha \mathrm{VaR}_t(u)\,\mathrm{d}u$$

- VaR is a **threshold**; ES is an **average of the tail**.
  - VaR locates the edge of the tail, ES averages what lies past it.
  - On a continuous law that average is $\mathbb{E}[L_t \mid L_t > \mathrm{VaR}_t]$.
- We score **VaR** today: every model family here can be asked for it.
  - ES returns in the appendix, where scoring it needs the pair.
- Throughout: VaR and ES are **positive magnitudes**.

Where do VaR and ES actually sit?





Two objects, and only one of them is a threshold

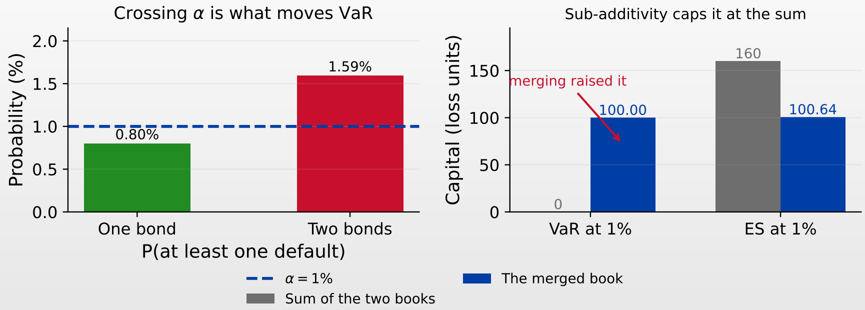
- The **edge** of the shaded band is the VaR; the **mean** inside it is the ES.
- Here that edge sits at -3.45 and the tail mean at -5.01, so VaR $= 3.45$ and ES $= 5.01$.
- Right: a Gaussian and a Student-t_3, each scaled to that same VaR.
- Their expected shortfalls are 3.95 and 5.32, so VaR alone reports the same risk for both.

What does coherence ask of a risk measure?

- $\rho(L)$ is the **capital** a book with loss L must hold.
  - A measure with all four properties below is **coherent**.
- **Monotonicity.** $L_1 \le L_2$ always $\Rightarrow \rho(L_1) \le \rho(L_2)$.
  - More loss, more capital.
- **Translation.** $\rho(L + c) = \rho(L) + c$ for a certain loss c.
  - Cash set aside counts once, at face value.
- **Homogeneity.** $\rho(\lambda L) = \lambda\, \rho(L)$ for $\lambda \ge 0$.
  - Double the book, double the requirement.
- **Sub-additivity.** $\rho(L_1 + L_2) \le \rho(L_1) + \rho(L_2)$.
  - Diversification can only lower the requirement, and this is the one VaR loses.

- VaR satisfies the first three (Artzner et al., 1999).
- It is sub-additive on **elliptical** losses: jointly Gaussian, or Student-t with a common ν.
- Outside it VaR can fail, and Basel moved capital to 97.5% ES (Basel Committee on Banking Supervision, 2019).

Two bonds, and the axiom VaR loses



Merging two books raised the requirement

- Held alone, each default sits *beyond* the 1% quantile: VaR reports 0 on both books.
- Merged, $P(\text{at least one}) = 1.59\%$ crosses α, so VaR becomes 100 — while ES gives $80 + 80$ against 100.64 and holds.

Two factors, but only one is richly identified

- σ_t is the **scale**, $q_z(\alpha)$ the quantile of the standardised shock z_t.

$$q_t(\alpha) = \mu_t + \sigma_t \, q_z(\alpha) \qquad \text{when} \qquad r_t = \mu_t + \sigma_t \, z_t, \quad z_t \text{ i.i.d.}$$

  - A *location–scale* law: the shape is the same on every day. Every classical benchmark assumes it.

The scale

- Re-estimated every day from the recent past.
- Volatility clusters, so it is forecastable.
- Thousands of examples to learn from.

The shape

- One observation per hundred days.
- A year of data holds **two or three**.
- The tail shape rests on those few.

- 1% is not 5% with a smaller number.
  - It is the same question asked where the data runs out, and each method degrades differently on the way.
- Every classical result today says which of the two it got wrong.

Four questions people ask of one table

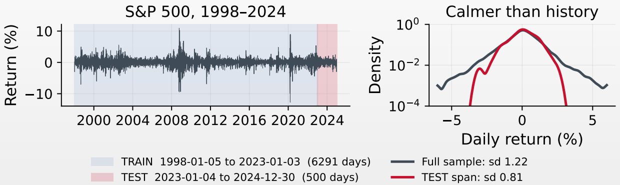
- **PIPELINE AUDIT** — did the object you scored come back from the question you asked?
  - Prior to the other three, and the one Act III is about.
- **RANKING** — which forecast has the lower consistent loss?
  - A relative statement: it orders two forecasts and stops there.
- **VALIDITY** — is the forecast compatible with the target coverage, and with the dynamics of the breaches?
  - An absolute test against a null, one model at a time.
- **POWER** — would that test have detected a materially invalid model on this sample?
  - A property of the experiment, fixed before any model runs.
  - Reported far less often than the other three.

- A frame that answers one of these carries its label in the top-right corner. Frames that only show a mechanism carry none.

Act II

The classical answers

The data



- **TRAIN** is a rolling window, re-estimated as it goes.
  - **1000** days long, refit every 20, using data up to $t-1$.
  - Burn-in: the recursion needs a pre-sample σ_0^2, so **250** days are dropped as its influence decays.
- That window belongs to the models that *estimate*. A pretrained model arrives fitted, so it gets a *context* instead.
- **TEST** is held out for scoring, with a headline archive over it.
  - Calm days: sd 0.81 against 1.22, excess kurtosis 0.7 against 9.9.

AIDA_aug_data

The simplest answer: sort the window

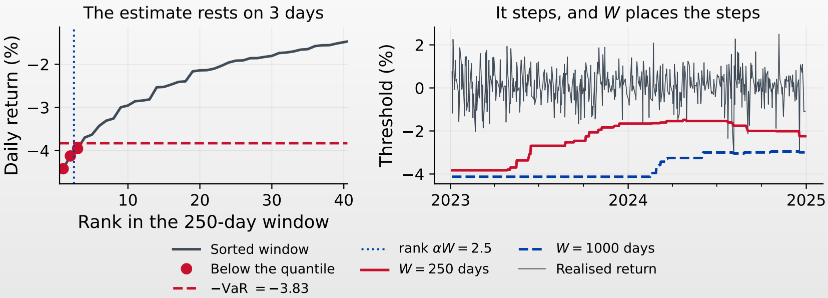
- Historical simulation: sort the last W days, read off the empirical quantile.

$$\widehat{\mathrm{VaR}}_t^{\mathrm{HS}}(\alpha) = -\,\widehat{Q}_\alpha(r_{t-W}, \ldots, r_{t-1})$$

  - The window supplies the tail shape directly.
- So the only modelling decisions left are W and the weighting.
  - Both are visible choices, and both move the answer.
- Every day in the window carries weight $1/W$.
  - At $W = 1000$, a day from four years ago counts as much as yesterday.
- It reacts only **discretely**, when a day enters or leaves the window.
  - So the response is coarse, and where it steps depends on W.

AIDA_aug_data

What historical simulation is actually doing



- In a 250-day window $\alpha W = 2.5$: with interpolation the estimate rests on the **three** worst days in it.
- Right: the same estimator at $W = 250$ and $W = 1000$, on the same days.
- It takes **21** distinct values over the span against **13**, from one estimator on two window lengths.

GARCH(1,1): model the scale, assume the shape

- One lag of the squared shock, one of the variance.

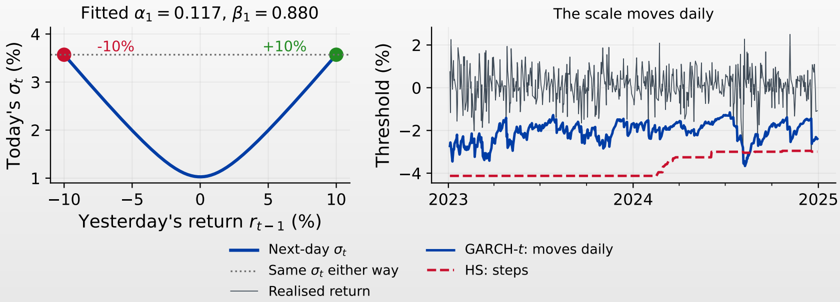
$$\varepsilon_t = r_t - \mu, \qquad \sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \qquad \widehat{\mathrm{VaR}}_t(\alpha) = -\left(\hat{\mu} + \hat{\sigma}_t\, q_{\hat{\nu}}(\alpha)\right)$$

  - α_1 weights yesterday's shock, β_1 yesterday's variance.
- The tail **family is imposed** on the standardised shock $z_t = \varepsilon_t / \sigma_t$; only its parameters are estimated.
  - Here it is **Student-**t.
  - A raw t_ν has variance $\nu/(\nu - 2)$, so it is rescaled to make $\mathrm{Var}(z_t) = 1$:

$$q_\nu(\alpha) = t_\nu^{-1}(\alpha)\sqrt{\tfrac{\nu-2}{\nu}}$$

  - At $\nu = 6.5$: $-3.06 \times 0.83 = -2.55$.
  - Skip the rescaling and VaR comes out **20%** too large.
- This is the scale/shape split written as one equation.
  - Published in 1982 and 1986 (Bollerslev, 1986; Engle, 1982), and still the benchmark to beat.

What the squared shock costs you



- Shocks of -10% and $+10\%$ leave the same σ_t, because the shock enters squared.
- Volatility usually reacts more to a fall than to an equal rise — the **leverage effect**, which sits outside this model.
- Right: the scale moves every day, where historical simulation holds still and steps.

Quantile regression: what goes in

- Fitted on **realised** pairs from the estimation window:
  - the return r_t, against predictors x_{t-1}.
- Seven columns:
  - r_{t-1}, $|r_{t-1}|$, the 22-day minimum, three **realised volatilities** and one **EWMA**.

$$\mathrm{RV}_t^{(w)} = \sqrt{\tfrac{1}{w}\sum_{i=1}^{w} r_{t-i}^2}\ \ (w = 5, 22, 66), \qquad \sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_{t-1}^2,\ \lambda = 0.94$$

  - All estimates of the same scale, equally against exponentially weighted.
- Choose a function $f(x_{t-1};\theta)$ mapping those predictors to one number.
  - Linear $f = \theta^\top x_{t-1}$, a tree ensemble, or a network.

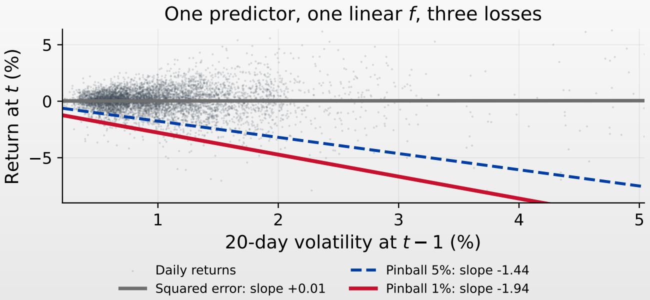
Quantile regression (QR): the loss decides what you fit

- Fit θ with the **pinball** loss (Koenker & Bassett, 1978) in place of squared error.

$$L_\alpha(r, q) = (\alpha - 1\{r < q\})\,(r - q), \qquad \hat{\theta} = \arg\min_{\theta} \sum_t L_\alpha(r_t, f(x_{t-1}; \theta))$$

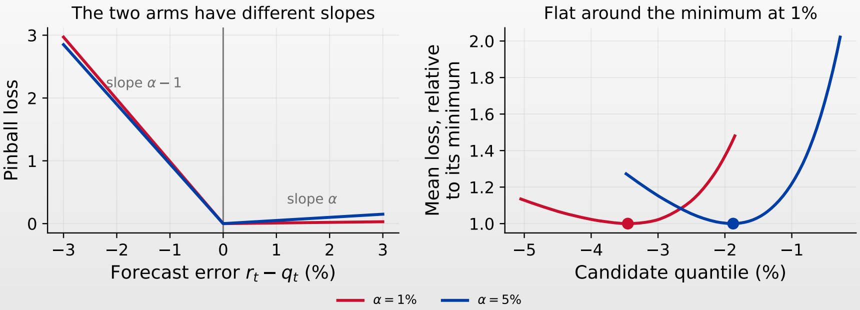
  - Squared error would place f at the conditional *mean*.
  - QR places it at the conditional *quantile*.
  - To forecast, feed it *today's* predictors: $\widehat{\mathrm{VaR}}_{t+1}(\alpha) = -f(x_t; \hat{\theta})$.
- The loss is **strictly consistent** for the quantile.
  - Minimised at the true quantile, so every ranking here uses it (Gneiting, 2011).
- The difficulty is the *level* you point it at.
  - Near 1% the surface is locally flat.
  - So f fits the centre while the tail drifts.

The loss moves the line



- One predictor here, for the picture: the 20-day volatility.
- Squared error puts the line at the mean: slope $+0.01$, flat and useless for risk.
- Pinball at 1% puts it at the quantile: slope -1.94, steep in volatility.

The pinball loss, and why 1% is the hard case



- Left: at $\alpha = 1\%$ the two arms have slopes 0.01 and -0.99.
- Right: the mean loss is flat around its minimum.
- A candidate can sit **0.64** points away and still score within one part in a hundred of the best — against 0.46 at $\alpha = 5\%$.

Where can AI actually help?

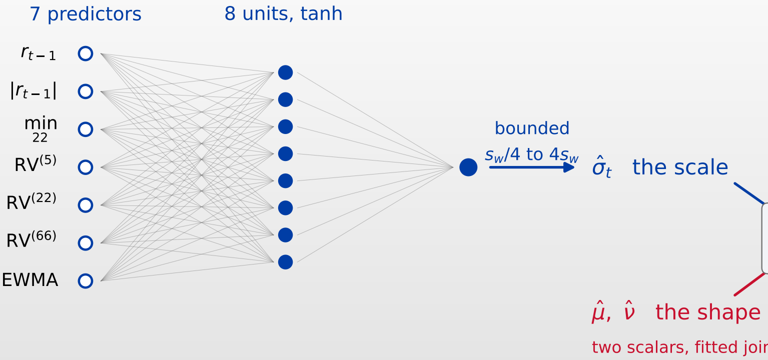
- **The scale** has thousands of training examples.
  - Every day is one, and volatility persists: an ordinary regression.
- **The shape** has a handful.
  - Two or three tail events a year, whatever the model.
  - The count belongs to the data, and every architecture faces it.
- So hand the learner the scale, and keep the shape parametric.
  - **NN-t** — a **neural network** for the scale, a Student-t for the tail. It predicts $\log \sigma_t$ from the same seven predictors, and is scored in every table from here on.

$$\sigma_t = \exp g(x_{t-1};\theta), \qquad \hat{\theta} = \arg\max_{\theta} \sum_t \left[\log f_\nu\left(\frac{r_t}{\sigma_t}\right) - \log \sigma_t\right], \qquad \widehat{\mathrm{VaR}}_t(\alpha) = -\hat{\sigma}_t\, q_{\hat{\nu}}(\alpha)$$

  - GARCH's own likelihood, with the variance *learned* rather than imposed by a recursion.

- A pretrained model claims *both* factors at once, from a corpus of other series.
  - Any gain must come from transferable structure, extra information or an assumption, and each of the three is checkable.

NN-t: eight units for the scale, two scalars for the tail



$$\hat{q}_t(\alpha) = \hat{\mu} + \hat{\sigma}_t\, q_{\hat{\nu}}(\alpha)$$

☐ $\hat{\sigma}_t$ is bounded between $s_w/4$ and $4s_w$, with s_w the estimation window's standard deviation.

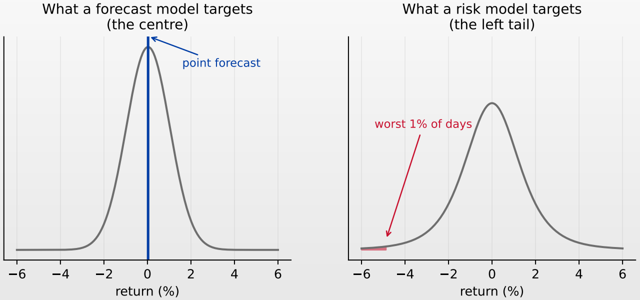
Act III

Time-series foundation models

Four words people use interchangeably

- **Supervised forecasting** — fit on *this* series, predict it.
- **Pretraining** — fit once, on a large corpus of *other* series.
- **Fine-tuning** — take a pretrained model, continue training on yours.
- **Zero-shot** — take a pretrained model and run it as it arrives.
  - **Zero-shot** covers Chronos and every language model, Acts III and IV.
  - **HS, GARCH-t and NN-t** are the supervised ones, fitted on this series.

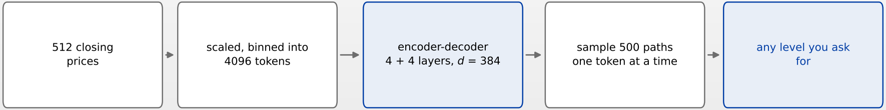
Three outputs, and only one of them is a VaR



- Three outputs get reported as if they were interchangeable.
  - A **point** forecast, **marginal quantiles** on a fixed grid, or a **predictive law**.
- VaR needs the second *at your level*, or the third.
  - A model that gives you quantiles on **its** grid has given you its grid.

Chronos-T5

Chronos-T5 mini, pretrained elsewhere



every parameter arrives fitted on other series

- The weights arrive trained and stay fixed.
  - It samples paths, so a quantile at any level is an order statistic of the draws.

AIDA_aug_chronos

Chronos: the series as a language

- Values are scaled and quantised into a fixed vocabulary of tokens.
  - So the output lives on that grid, which is where the next two frames start.
- A sequence model is pretrained on a large corpus of time series.
  - That corpus excludes this asset, and its full contents stay closed to inspection.
- Forecasting is next-token prediction, decoded back to the value scale.
  - The quantile you get is whatever the decoder can express.

- **Chronos-T5** samples paths, so any quantile can be read off (Ansari et al., 2024).
- TimesFM (Das et al., 2024) is a different architecture on the same idea.

AIDA_aug_chronos

Chronos-T5 forecasts the next value, so we hand it prices

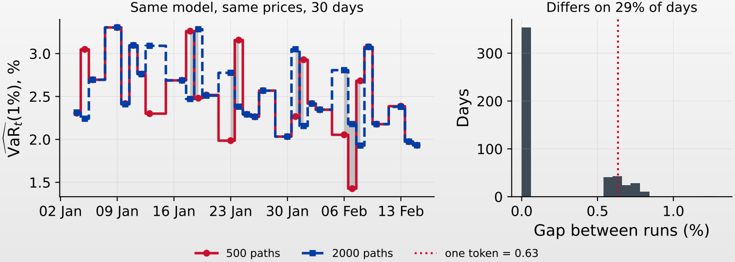
- Chronos-T5 was trained on series **levels**, and rescales each context by its own mean and spread.
  - So it gets a *context* of the last 512 closing prices, the length it was trained to read, and returns a distribution for the next.
- That distribution is over prices, and we need one over returns.
  - Each sampled price $\hat{P}_t$ becomes a return against the last observed close P_{t-1}: $\hat{r}_t = 100 \log(\hat{P}_t / P_{t-1})$.
- That step preserves the quantile, because the map is increasing in $\hat{P}_t$.
  - The α quantile of the price is therefore the α quantile of the return.

What happens if you feed returns instead

- A return series is already near zero mean with a small spread, so the rescaling has nothing left to do.
  - The predictive spread collapses, and the VaR collapses with it.

The same model, the same day, two different VaRs

PIPELINE AUDIT



- Same weights, same prices, same dates. Only the number of sampled paths differs.
- The two estimates part on **29%** of days, and when they part it is by one token, **0.63** percentage points.
- At the library default of 20 paths, the lowest draw already sits above the 1% quantile.

AIDA_aug_chronos

Act IV

Language models as risk forecasters

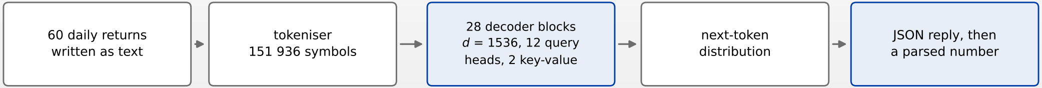
Five different things people mean by "LLMs for risk"

1. **Feature extraction** — sentiment from text, fed to a conventional model.
   - The AI stops at a feature. The risk number is still econometrics.
2. **Event classification** — label the news flow.
   - Still a feature: the risk model consumes the label.
3. **Probability elicitation** — ask for $\mathbb{P}$(tail event).
   - A number in $[0, 1]$, which needs a threshold before it is a quantile.
4. **Quantile elicitation** — ask for the VaR itself.
   - The model states the threshold, and checking that it is one is left to you.
5. **Predictive simulation** — sample tomorrow's *return*, many times.
   - The draws are a distribution; the quantile is computed from it.

- The last three put the large language model (LLM) *inside* the risk number.
  - **Quantile elicitation**: Claude Haiku 4.5 in four configurations, and Qwen2.5-1.5B.
  - **Predictive simulation**: Qwen2.5-1.5B at 1000 draws (Pele et al., 2026).

A language model reaches a quantile through text

Qwen2.5-1.5B-Instruct, decoder-only



Every layer is general purpose. The request makes it a quantile.

- One tokeniser, 28 decoder blocks and one next-token head, all shared with prose generation.
- The 1% level enters through the request and leaves through a parsed JSON field.

AIDA_aug_local_llm

Two ways to get a quantile out of a language model

- Both begin by **serialising**: the return history is written out as text, two decimals and comma separated, after Gruver et al. (2023).

Predictive simulation — many draws

- The quantile is an order statistic of the draws, so N sets how fine it can be (Pele et al. (2026)).
  - Run here on Qwen2.5-1.5B at $N = 1000$.

Quantile elicitation — one call instead of N

- The model is asked for q_α in words and the number is parsed out of the reply.
  - Run on Claude Haiku 4.5 and Qwen2.5-1.5B.

AIDA_aug_llm

The prompt is a modelling decision

- Four information sets, and the differences between them are the experiment:
- `series` — the last 60 returns, **anonymised**.
  - Stripped of ticker, dates and price levels.
- `series+state` — the same, plus volatility, drawdown, the HS VaR.
- `dated` — the same returns, plus the asset name and the date.
- `dated+news` — the same, plus the real headlines available at $t-1$.

Why two of them hide the date

- The weights were trained on text that already covers 2023 and 2024.
  - Given "S&P 500, 5 August 2024", the model can *remember* what happened instead of forecasting it.
  - A live forecast has only the past to work with, so a score won that way would not survive live use.

The sampling prompt

PIPELINE AUDIT

```
SYSTEM
You are a helpful assistant that performs time series predictions. The
user will provide a sequence and you will predict the remaining sequence.
The sequence is represented by decimal strings separated by commas.

USER
Please continue the following sequence without producing any additional
text. Do not say anything like 'the next terms in the sequence are', just
return the numbers. Sequence:
```

```
0.89, -0.96, 0.96, 0.71, -0.21, 0.60, 0.77, -0.76, 0.47, -0.02, 0.40, -0.18, -0.05, -0.92, 0.21, -0.03, ...
```

The elicitation prompt

PIPELINE AUDIT

System

```
You are a market risk analyst. From a series of recent daily returns you estimate the risk
of the next trading day. You answer with a single JSON object and nothing else: no
explanation before it, no code fence around it.
```

User

```
Daily returns in percent, oldest first, 60 observations:
-2.84, -0.75, -0.65, -0.33, 2.56, -2.39, 2.61, 1.14, -0.67, -0.80, ...  ..., 1.73, -0.25, -0.40
```

Schema, appended to every configuration

```
Return exactly this JSON object:
{
  "q01": <number>,              // the 1% quantile of tomorrow's return, in percent. Negative.
  "q05": <number>,              // the 5% quantile of tomorrow's return, in percent. Negative.
  "prob_negative": <number>,   // probability tomorrow's return is below zero, 0 to 1
  "prob_tail": <number>,       // probability tomorrow's return is below -2%, 0 to 1
  "risk_score": <integer>,     // 0 calm to 100 extreme
  "confidence": <number>,      // your confidence in q01, 0 to 1
  "drivers": [<string>, ...]   // at most three short phrases
}

q01 must be more negative than q05: a 1% quantile lies further into the left tail.
```

Why the control is `dated`

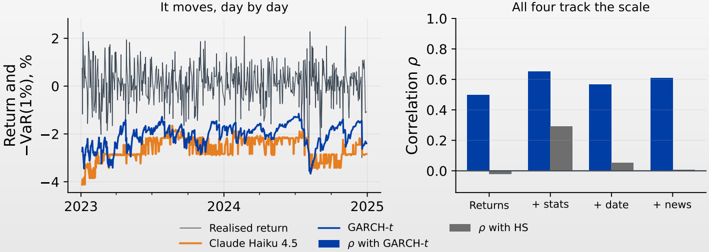
- A headline enters the prompt for day t only if it was published before that market's own close on $t-1$.
  - Consecutive cutoffs tile the calendar, so each article is counted exactly none is dropped.
- Real headlines also *reveal the period*.
  - They name companies, events and numbers that place the text in time.

Configuration	period revealed	headlines	role
`series`	no	no	anonymised baseline
`dated`	yes	no	**control**
`dated+news`	yes	yes	**treatment**

- Comparing `series` with a news configuration would confound "the text carries information" with "the model now knows which period this is".
  - Only the last two rows identify the first effect.

Claude states a threshold, and it moves

RANKING

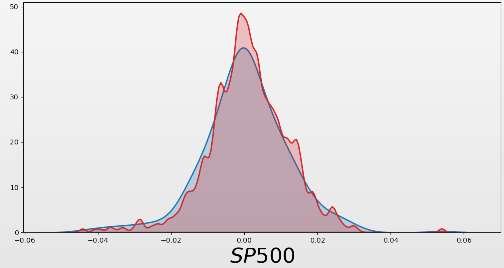


- The four bars are the four prompts: returns alone, then plus statistics, plus the date, plus headlines.
- Returns alone correlates **0.50** with GARCH-t, and all four correlate far more with it than with HS.
- Consistent with the model tracking volatility — the factor that has data.

AIDA_aug_llm

What the sampled distribution looks like next to the truth

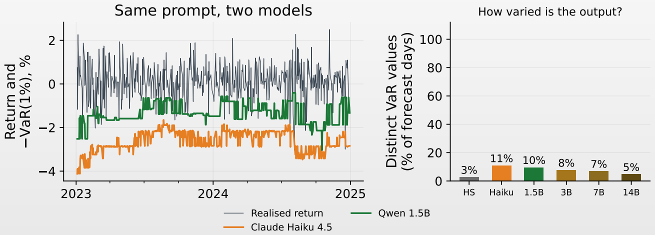
PIPELINE AUDIT



- **Blue**: realised returns. **Red**: the model's draws, pooled over the test span.
- The gap sits in the shoulders. A pooled *marginal* diagnostic, which leaves conditional calibration open. Pele et al. (2026), GPT-4.

Counting the distinct answers, and what it does not prove

PIPELINE AUDIT



- Right: the share of the 500 days carrying a *distinct* VaR.
  - Under 11% for every language model, and 2.6% **for historical simulation**, which passes coverage.
- So a low count opens a question about the mechanism.
  - HS repeats for a reason you can write down; the others repeat for one held in the weights.

AIDA_aug_local_llm

Seven ways to produce one quantile

Family	Input	Native output	VaR from it	Hidden failure	First check
Historical simulation	Last W returns	Empirical law	Order statistic	Coarse, window-dated moves	Window length
GARCH-t	Returns	σ_t and assumed shape	$-(\mu + \sigma_t q_\nu(\alpha))$	Tail family imposed	Shape assumption
Quantile regression	Features	The quantile itself	Direct	Flat objective near 1%	Identification
NN-t	Features	Learned σ_t, assumed shape	$-(\hat{\mu} + \hat{\sigma}_t q_{\hat{\nu}}(\alpha))$	Architecture picked on the sample	Selection slice
Chronos-T5	Price levels	Sample paths	Order statistic	Resolution floor at $1/N$	Draw count
LLM, elicited **Claude Haiku 4.5, Qwen2.5-1.5B**	Serialised text	A stated number	Parse the reply	Passes every check, barely moves	Distinct values
LLM, sampled **Qwen2.5-1.5B**	Serialised text	N completions	Order statistic	The same floor, plus parsing	Draws and parser

- Seven routes to the same number, the 1% VaR.
- Each breaks in a way that still returns a plausible number, so the backtest runs as usual.
- **First check** is what catches it beforehand, and every one of them runs before the scoring does.

Act V

Validation

Backtesting in one line

- Count the days the threshold was crossed:

$$\hat{\alpha} = \frac{1}{T}\sum_{t=1}^{T} 1\left\{r_t < -\widehat{\mathrm{VaR}}_t(\alpha)\right\}$$

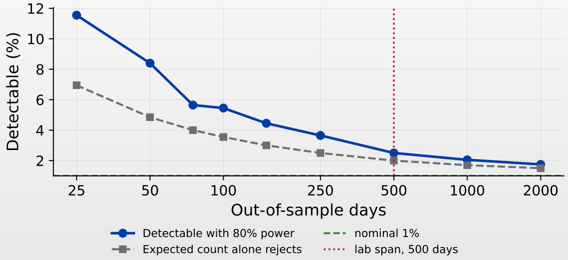
- **Kupiec** tests whether $\hat{\alpha}$ differs from α (Kupiec, 1995).
  - $LR_{UC} \sim \chi^2_1$ under the null.
- **Christoffersen** (Christoffersen, 1998) tests whether the breaches *cluster*.
  - Right count, wrong timing is still a failure: breaches that arrive in one fortnight are the fortnight that matters.
- Their sum is conditional coverage, χ^2_2.
  - The full expressions, and the Diebold–Mariano (DM) test that compares two models, are in the appendix.

▸ Kupiec and Christoffersen ▸ Diebold–Mariano ▸ Which loss

AIDA_aug_backtest

How wrong a model must be before 500 days can see it

POWER

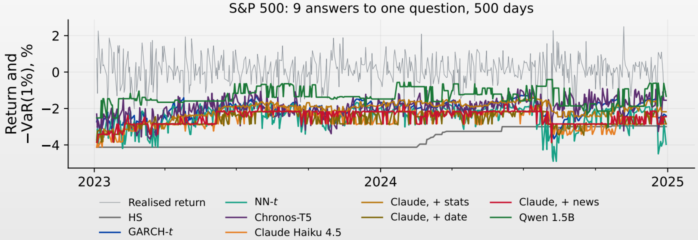


- Two-sided Kupiec LR_{UC} at 5%, exact binomial power.
- 80% power arrives at **2.50%** over 500 days.
- Over 5541 days, twenty-two years of daily data, it arrives at 1.45%.

AIDA_aug_backtest

Every forecast, on one axis

RANKING

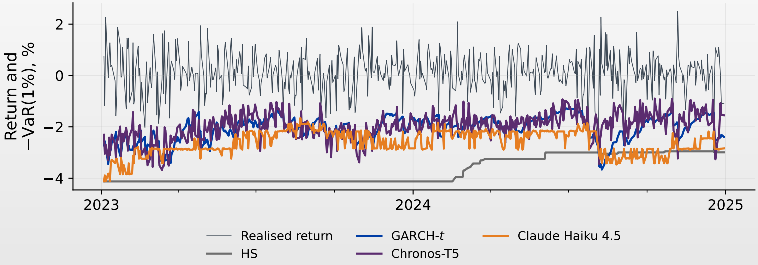


- Nine forecasts, the same 500 days, the same target.
  - Mean thresholds run from 1.35 to 3.69, a factor of nearly three.
- **Qwen sits closest to zero, and breaches seven times too often.**
  - Historical simulation is the staircase at the bottom, and is rejected on two markets for being too safe.

AIDA_aug_backtest

The same question, four answers

RANKING



- Four models, one plot. Each line is that model's 1% VaR for the day, drawn in return space.
- The history each sees is set by the method: 1000 days to estimate on, 512 because that is the context Chronos was trained to read, 60 of text.
- **The day, the cutoff at $t-1$ and the scoring are identical.** That is what makes the four comparable.

What the backtest says

VALIDITY

Model	n	Breaches	Rate (%)	UC	IND	Pinball	DM vs GARCH
Neural volatility + t tail	500	3	0.60	pass	pass	0.0269	0.69
Claude Haiku 4.5, dated + headlines	500	3	0.60	pass	pass	0.0276	0.95
GARCH(1,1)-t	500	7	1.40	pass	pass	0.0277	*reference*
Claude Haiku 4.5, series + state	500	6	1.20	pass	pass	0.0291	0.60
Claude Haiku 4.5, dated (control)	500	3	0.60	pass	pass	0.0298	0.23
Claude Haiku 4.5, series only	500	3	0.60	pass	pass	0.0299	0.25
Chronos-T5, 500 sample paths	500	6	1.20	pass	pass	0.0303	0.31
Historical simulation	500	2	0.40	pass	pass	0.0379	**0.004**
Qwen2.5-1.5B, sampled 1000×	500	**25**	**5.00**	reject	pass	0.0436	**0.006**
Qwen2.5-1.5B, asked	494	**35**	**7.09**	reject	pass	0.0533	**0.013**

- Sorted by pinball loss; **UC** is Kupiec, **IND** is Christoffersen, both at 5%.
- Five breaches were expected in 500 days; Kupiec rejects only the two Qwen rows.
- **The top three**: NN-t, Claude with headlines and GARCH-t.
- **DM** is the Diebold–Mariano p-value against GARCH-t: it separates three of the nine, and all three score worse.

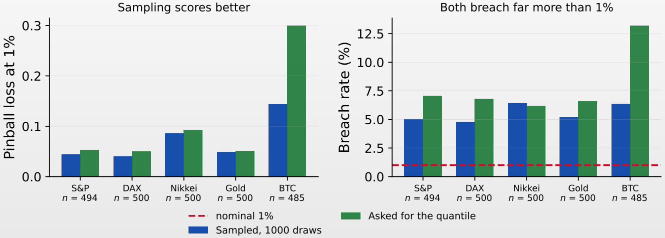
Eight of ten pass, and that says more about the test

POWER

- Breach counts of 2, 3, 6 and 7 all pass.
  - With five events expected, all four counts sit inside the acceptance region.
- The two rejections had to miss by a factor of five and of seven.
  - Realised rates of 5.00% and 7.09% against a nominal 1%.
- So eight passes tell you the sample is too short to separate them.
  - **Report the power beside the verdict, or the verdict is decoration.**

The same weights, asked two different ways

RANKING

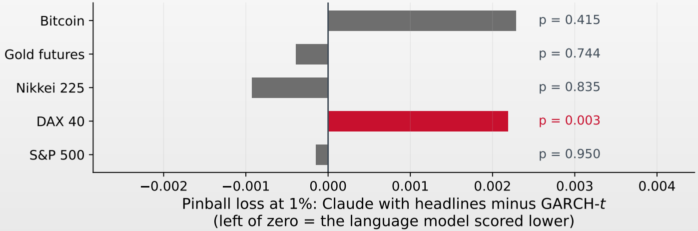


- Sampling scores the lower loss on all five markets, at 1000× the compute.
  - One model, Qwen2.5-1.5B, two ways of asking, same days and same parser.
- **Both breach far too often to be a 1% forecast.**
  - Kupiec rejects all ten runs, at 4.8% to 13.2%.

AIDA_aug_local_llm

Does the language model match the benchmark?

RANKING



- **On four of five markets the two score alike.**
  - On the DAX, GARCH-t is better at $p = 0.003$.
- Matching the benchmark is the best result on offer here.
  - On the remaining four markets the difference stays inside sampling noise.

AIDA_aug_backtest

Does the S&P result replicate?

VALIDITY

Model	S&P 500	DAX 40	Nikkei 225	Gold futures	Bitcoin
Neural volatility + t tail	3 (0.6%)	3 (0.6%)	4 (0.8%)	5 (1.0%)	2 (0.4%)
GARCH(1,1)-t	7 (1.4%)	3 (0.6%)	6 (1.2%)	7 (1.4%)	**1 (0.2%)**
Historical simulation	2 (0.4%)	**0 (0.0%)**	5 (1.0%)	7 (1.4%)	**0 (0.0%)**
Chronos-T5, 500 sample paths	6 (1.2%)	**11 (2.2%)**	**12 (2.4%)**	**16 (3.2%)**	**13 (2.6%)**
Claude Haiku 4.5, dated + headlines	3 (0.6%)	5 (1.0%)	3 (0.6%)	7 (1.4%)	4 (0.8%)
Claude Haiku 4.5, series + state	6 (1.2%)	7 (1.4%)	7 (1.4%)	**11 (2.2%)**	7 (1.4%)
Claude Haiku 4.5, dated (control)	3 (0.6%)	4 (0.8%)	3 (0.6%)	4 (0.8%)	5 (1.0%)
Claude Haiku 4.5, series only	3 (0.6%)	3 (0.6%)	3 (0.6%)	2 (0.4%)	8 (1.6%)
Qwen2.5-1.5B, asked	**35 (7.1%)**	**34 (6.8%)**	**31 (6.2%)**	**33 (6.6%)**	**64 (13.2%)**
Qwen2.5-1.5B, sampled 1000×	**25 (5.0%)**	**24 (4.8%)**	**32 (6.4%)**	**26 (5.2%)**	**31 (6.2%)**

- Breaches, with the realised rate in brackets. **Red** is a Kupiec rejection at 5%.
- **18 of the 50** cells are rejected: both Qwen rows everywhere, Chronos-T5 on four markets of five.
- Every cell rests on 500 scored days, except where structurally invalid replies were dropped. Each test uses its own n.

AIDA_aug_backtest

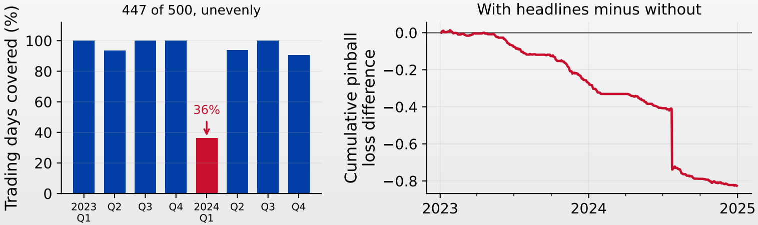
Reading that table without fooling yourself

POWER

- Fifty tests at the 5% level.
  - Chance alone would give two or three rejections. There are **18**.
- They fall on three of the ten rows.
  - Ten are the two open-weights rows; four more are Chronos-T5.
- Chronos-T5 is what the single-asset table hid.
  - 1.2% on the S&P, then 2.2, 2.4, 3.2 and 2.6% elsewhere.
- Historical simulation fails the other way.
  - Zero breaches on the DAX and Bitcoin.
  - **Too conservative is a rejection too**, at $p = 0.0015$.

Real news, at the level where it bites

RANKING

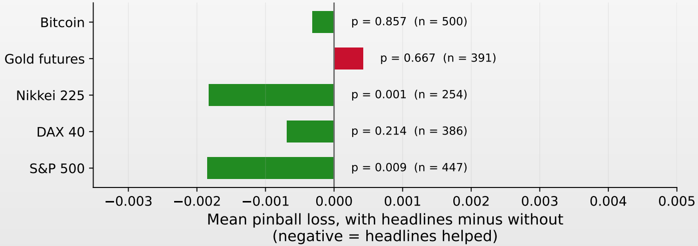


- **Both runs breach on the same 3 days, so coverage is silent here.**
  - $p = 0.331$ for each, against the same nominal 1%.
- The loss uses all **447** covered days and falls **6.5%**.
  - HAC (autocorrelation-robust) 95% CI $[1.6\%, 11.3\%]$, but one day carries **39%** of it.

AIDA_aug_news

Does it hold on the other four markets?

RANKING



- Significant on the S&P and the Nikkei; the other three sit inside noise.
  - Five markets is five tests, so **Bonferroni** sets the bar at 0.01. Both clear it.
- **Headlines help this model, on these markets, at this level.**
  - The claim stops at these five markets and this level.

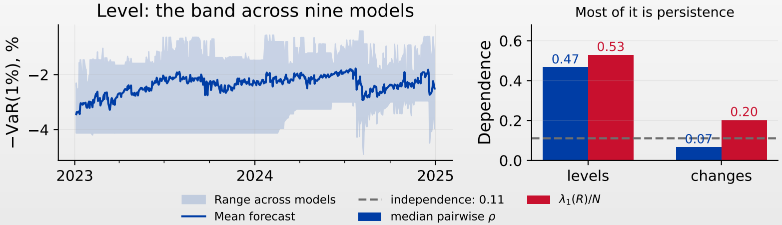
AIDA_aug_news

Leakage: the failure that flatters every model

- **Window** leakage: the forecast day sits inside its own estimation window.
  - Every window here ends at $t-1$, the 250-day one included.
- **Alignment** leakage: the return series shifted one row against the dates.
  - Audited on the news merge, where the two feeds keep different calendars.
- **Selection** leakage: the architecture chosen on the period you report.
  - NN-t was sized on an early slice and left alone afterwards.
- **Temporal contamination**: the pretraining corpus already holds the outcome.
  - The prompt therefore arrives stripped of ticker, dates and price levels.
- The first three you can test for.
  - **The fourth stays open, because the corpus is closed.**

How much of the disagreement is real?

VALIDITY



- On *levels* the nine agree, on *changes* they barely do.
  - $\rho = 0.47$ and 0.53 against 0.07 and 0.20, with 0.11 under independence.
- **So the agreement is shared persistence.**
  - Their daily updates are close to unrelated, and the band spans 2.59pp.

AIDA_aug_backtest

Can AI produce valid risk forecasts?

YES

- **NN-t**, fitted.
  - Passes coverage on all five markets.
- **Claude Haiku 4.5**, prompted.
  - Three of its four configurations pass on all five.

NO

- **Chronos-T5**, pretrained on series.
  - Rejected on four markets of five, at 2.2% to 3.2%.
- **Qwen2.5-1.5B**, open weights.
  - Rejected on every market, at 4.8% to 13.2%.

- **Every AI failure here is the same failure**: a threshold too close to zero.
  - **15** of the 18 rejections are too many breaches.
  - The three too-few belong to historical simulation and GARCH-t.
- The two that pass are level with the benchmark.
  - On loss, the Diebold–Mariano test puts them level with GARCH-t.
  - On coverage, Kupiec rejects reliably only past a realised rate of 2.50%.

The laboratory

Two hours in one notebook, on your own laptop

What you will do this afternoon

- **Write two forecasts yourself, audit an artefact, then score the whole panel.**
  - One notebook, in Colab, on your own laptop, running on open weights.

1. Historical simulation you complete, and the power calculation for 500 days.
   - Seconds per day, checked against the reference implementation.
2. The Chronos-T5 audit: how much the forecast moves between two identical runs.
   - The check the lecture argued comes before any scoring.
3. The language model live, on 5 to 20 dates, with the parser and its checks.
   - Then the news comparison against its `dated` control.
4. Backtests, Diebold–Mariano and disagreement, on the full precomputed panel.
   - Five assets, ten models, 500 days, all precomputed.

AIDA_Risk_Lab.ipynb — open it in Colab

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Appendix

Derivations, diagnostics and the conventions this course assumes

What produced every number in this deck

Artefacts

- `amazon/chronos-t5-mini`, 500 paths
- `Qwen/Qwen2.5-{1.5,3,7,14}B-Instruct`, `float16`
- `claude-haiku-4-5`, temperature 1.0, 400 max tokens

Environments

- `chronos-forecasting` 2.2.2, `transformers` 4.57.6, `torch` 2.12.0
- Open weights: `transformers` 5.14.1, `torch` 2.13.0, Python 3.12.13, device `mps`
- `SEED = 42`; three pinned `requirements` files in the repo

- Every result here is a claim about *these* checkpoints under *these* library versions, and would have to be re-established under another.

One page of conventions this course assumes

Signs and objects

- r_t is the log return in %; the loss is $L_t = -r_t$.
- $q_t(\alpha) = F_t^{-1}(\alpha) < 0$ is the return quantile.
- $\mathrm{VaR}_t(\alpha) = -q_t(\alpha) > 0$: a positive magnitude.
- $\mathrm{ES}_t(\alpha) = \frac{1}{\alpha}\int_0^\alpha \mathrm{VaR}_t(u)\,\mathrm{d}u$: the worst α averaged.
- Breach: $I_t = 1\{r_t < -\widehat{\mathrm{VaR}}_t(\alpha)\}$.
- A **hat** marks a quantity computed from data.

Scale and shape

- $q_t(\alpha) = \mu_t + \sigma_t\, q_z(\alpha)$ under a location–scale law: scale re-estimated daily, shape held fixed.

The pinball loss

- $L_\alpha(r, q) = (\alpha - 1\{r < q\})(r - q)$.
- Minimised at the true quantile, so it ranks forecasts.
- At $\alpha = 1\%$ the arms have slopes 0.01 and -0.99.

How many breaches to expect at 1%

- 250 days: 2.5 expected; 500 days: 5.
- Which is why Act V spends time on power.

What makes a loss the right loss?

back

Definition 1 (Consistency)

A scoring function L is **consistent** for a functional T if $\mathbb{E}[L(Y, T(F))] \leq \mathbb{E}[L(Y, x)]$ for every F and every report x. It is **strictly** consistent if equality forces $x = T(F)$.

Definition 2 (Elicitability)

T is **elicitable** if some strictly consistent L exists for it. Then $T(F) = \arg\min_x \mathbb{E}[L(Y, x)]$, and models can be ranked by average loss.

- Quantiles are elicitable; the pinball loss does it (Gneiting, 2011).
- ES *lacks* one of its own, which Act V takes up.

What the feed gave us

EOD Historical Data, S&P 500 index feed

- 35 293 articles fetched
- 2 128 survive the as-of rule
- 447 of 500 dates covered (89%)
- 4.8 headlines per covered date
- **0** published after their cutoff

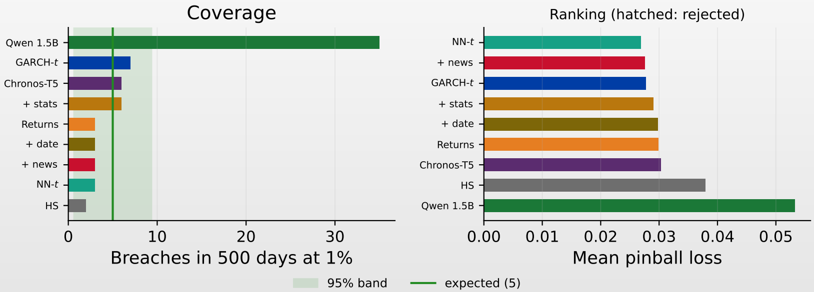
The first symbol was wrong

- `SPY.US` returned 222 articles for two years.
- It left **67%** of dates empty.
- The feed was chosen on measured coverage.

- Median publication lead before the cutoff: **1.1 hours**.
  - The text is genuinely fresh, and genuinely prior.

Coverage and ranking, side by side

RANKING



- Expected 5 breaches.
- Historical simulation delivers 2, GARCH-t 7, the open-weights model **35**.

AIDA_aug_backtest

Kupiec and Christoffersen in full

back

- With T days, x breaches and n_{ij} transitions between breach and no-breach:

Definition 3 (Unconditional coverage)

$$LR_{UC} = -2\log\frac{(1-\alpha)^{T-x}\,\alpha^{x}}{\left(1-\frac{x}{T}\right)^{T-x}\left(\frac{x}{T}\right)^{x}} \sim \chi^2_1$$

(Kupiec, 1995)

Definition 4 (Independence)

$$LR_{IND} = -2\log\frac{(1-\hat{\pi})^{n_{00}+n_{10}}\,\hat{\pi}^{\,n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}}\,\hat{\pi}_{01}^{n_{01}}\,(1-\hat{\pi}_{11})^{n_{10}}\,\hat{\pi}_{11}^{n_{11}}} \sim \chi^2_1$$

(Christoffersen, 1998)

- $\hat{\pi}_{i1} = n_{i1}/(n_{i0}+n_{i1})$ is the breach probability after state i; $\hat{\pi}$ pools them.
  - Conditional coverage is the sum of the two, χ^2_2.

Comparing two models on the same days

back

Definition 5 (Diebold–Mariano)

With $d_t = L_\alpha(r_t, q_t^A) - L_\alpha(r_t, q_t^B)$,

$$DM = \frac{\bar{d}}{\sqrt{\hat{\Omega}/T}} \xrightarrow{d} \mathcal{N}(0,1), \qquad H_0 : \mathbb{E}[d_t] = 0$$

(Diebold & Mariano, 1995)

- $\hat{\Omega}$ is a **HAC** estimate (Newey & West, 1987).
  - Loss differentials are serially dependent, so the naive variance can overstate significance.
- The test is paired, so both models must be scored on the *same* days.
  - A model that skips days is not comparable until the sample is aligned.
- Report $\bar{d}$ and its interval alongside the verdict.

Scoring ES means scoring the pair

Definition 6 (Fissler–Ziegel FZ_0 loss)

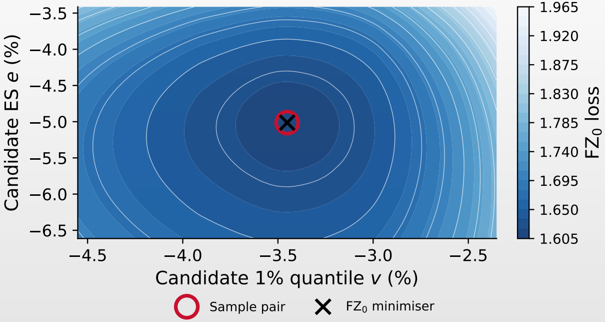
For a quantile v and a shortfall e with $e < v < 0$ at level α,

$$L_{\mathrm{FZ}_0}(r, v, e) = -\frac{1}{\alpha e} 1\{r \leq v\}(v - r) + \frac{v}{e} + \log(-e) - 1$$

(Fissler & Ziegel, 2016; Patton et al., 2019)

- ES **lacks a scoring function of its own**.
  - Every candidate loss is minimised somewhere else, so ranking on ES alone stays out of reach.
- The *pair* $(\mathrm{VaR}, \mathrm{ES})$ is jointly elicitable.
  - Which is why regulation can still be backtested (Nolde & Ziegel, 2017).

The minimum sits at the truth in both coordinates



- Full sample 1998–2024, so the pair is unconditional: the FZ_0 minimiser lands on $(-3.45, -5.01)$.
- The valley is far shallower along ES — the tail mean rests on 68 days of 6791.

SYNCRISK: when every desk runs the same model

$$D_t = \max_i \widehat{\mathrm{VaR}}_{i,t} - \min_i \widehat{\mathrm{VaR}}_{i,t}, \qquad \rho_{ij} = \mathrm{corr}\left(\widehat{\mathrm{VaR}}_i, \widehat{\mathrm{VaR}}_j\right)$$

- If every desk runs a similar model, every desk de-risks the same morning.
- Shared **pretraining** is a new channel for that.
  - The same weights behind many vendors, invisible from outside.
- Measured on **changes**: $\lambda_1/N = 0.20$ against 0.11.
  - Above independence, and far below the 0.53 a level correlation reports. The forecasts are persistent; correlating levels reads trend as agreement.
- Measuring ρ_{ij} across your own model set is a supervisory question.
  - You compute both quantities this afternoon.